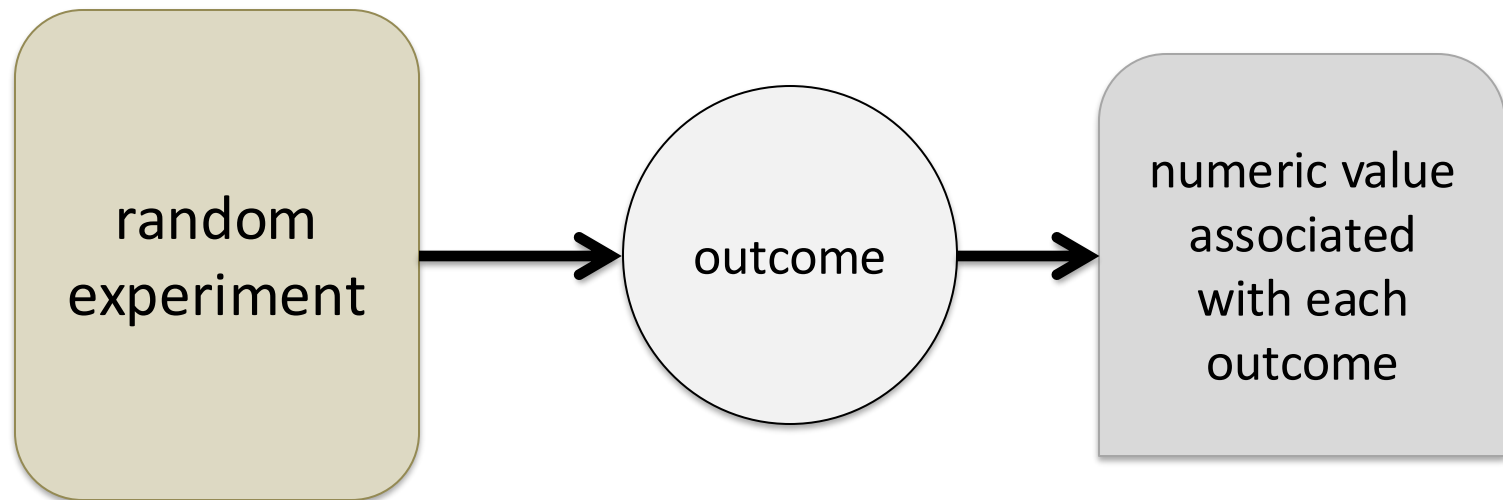


IEOR 4404: Simulation

Probability Review (con'd)

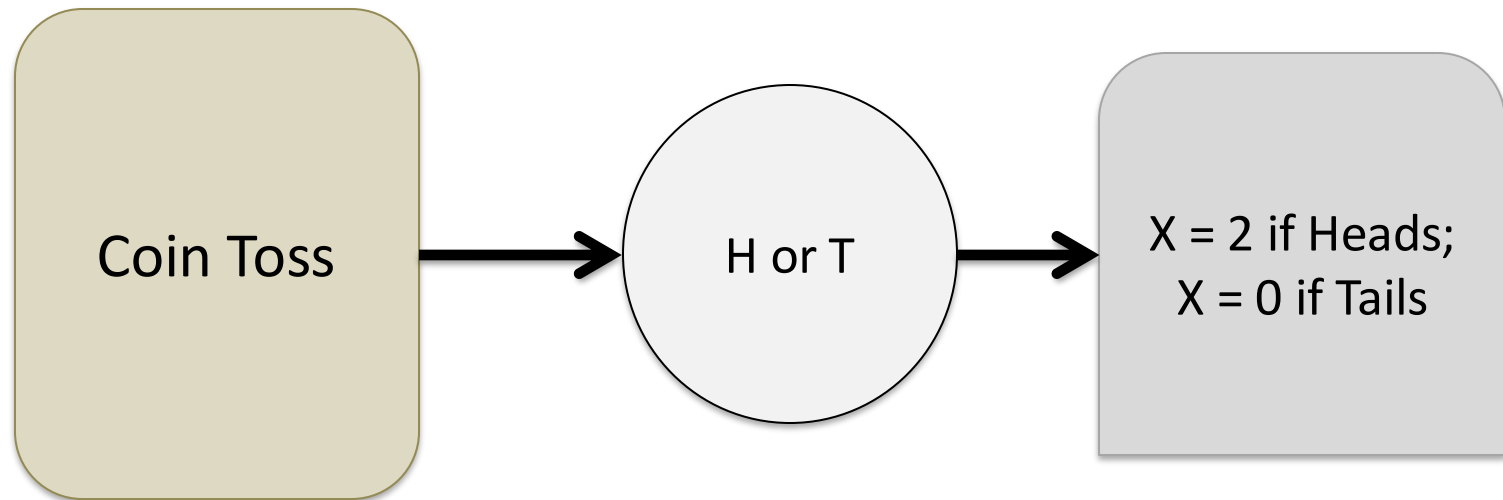
Random Variables

- Random variable summarizes or represents the outcomes with numerical values
- The realized value of a random variable is random
- To specify a random variable, identify its range of values and probability assigned to each value



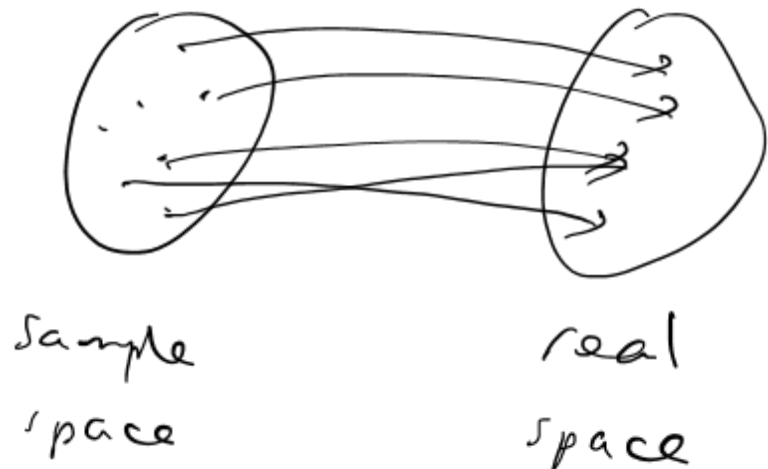
Random Variables

- Random variable summarizes or represents the outcomes with numerical values
- The realized value of a random variable is random
- To specify a random variable, identify its range of values and probability assigned to each value



Random Variables

- More formally: a *random variable* is a *function* that assigns a numerical value to each outcome in the sample space
- A random variable is *discrete* if the range of values that it takes is finite or “can be enumerated”
- Possible range of values:
 - $\{0, 1\}$ (coin toss example)
 - $\{0.5, 2, \pi, e\}$
 - $\{0, 1, 2, 3, 4, \dots\}$



Random Variables

- *Probability mass function* (pmf) $p(a)$ of a discrete random variable X is $p(a) = P(X = a)$, for each possible value a of X
- $p(a)$ can be positive for at most a *countable* number of values of a – if X takes values $x_1, x_2, x_3, \dots$, then $p(x_i) > 0$, $i = 1, 2, \dots$ and $p(x) = 0$ for all other values of x
- **Example.** Suppose a company hired 6 employees recently and let X be the number of females in the group. Further suppose that $P(X = 1) = 0.1$, $P(X = 2) = 0.3$, $P(X = 3) = 0.5$, $P(X = 4) = 0.05$, $P(X = 5) = P(X = 6) = 0.025$

pmf :

$$p(1) = 0.1$$

$$p(2) = 0.3$$

$$p(3) = 0.5$$

$$p(4) = 0.05$$

$$p(5) = p(6) = 0.025$$

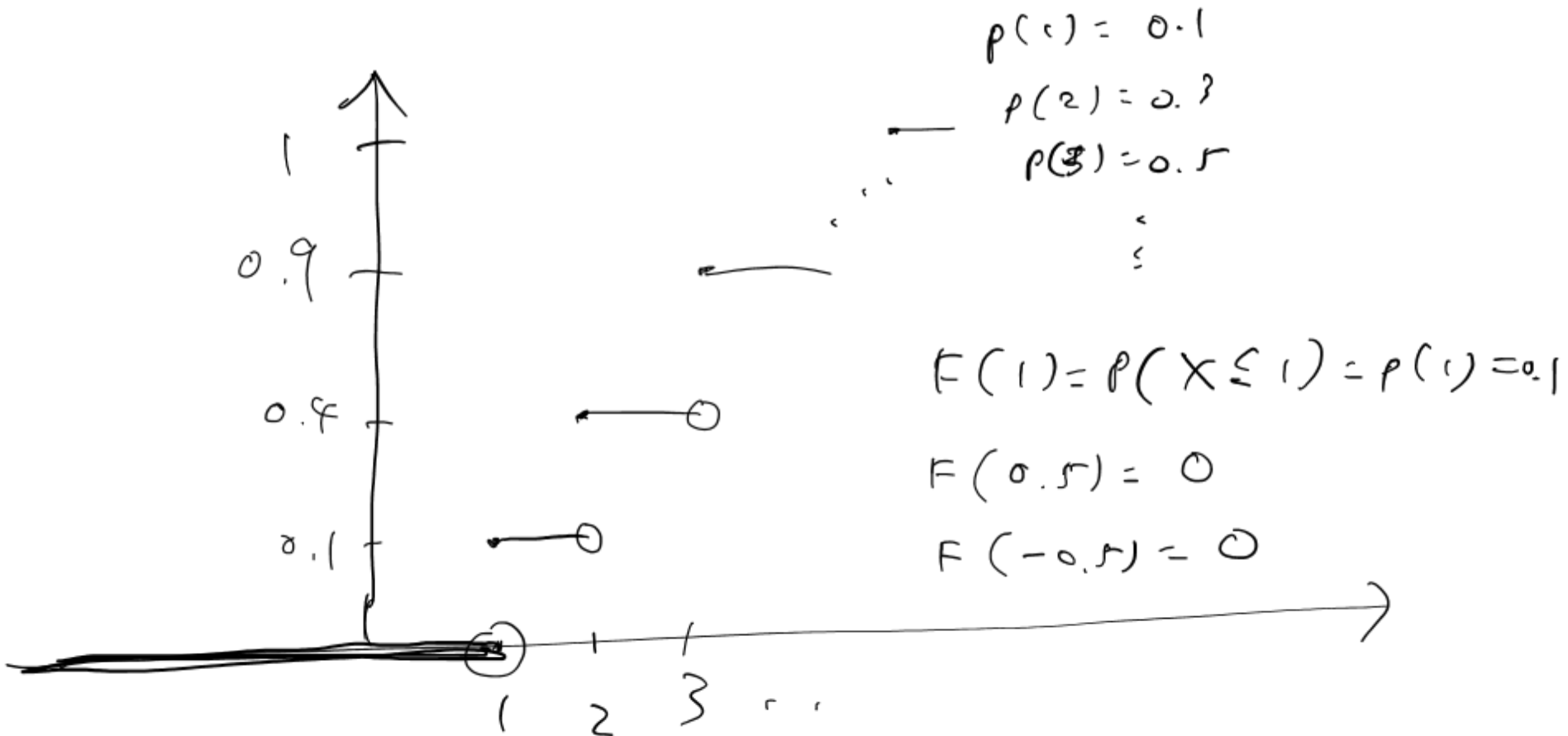
$$p(a) = 0 \text{ for all other } a.$$

Random Variables

- *Probability mass function* (pmf) $p(a)$ of a discrete random variable X is $p(a) = P(X = a)$, for each possible value a of X
- $p(a)$ can be positive for at most a *countable* number of values of a – if X takes values $x_1, x_2, x_3, \dots$, then $p(x_i) > 0$, $i = 1, 2, \dots$ and $p(x) = 0$ for all other values of x
- **Example.** Suppose a company hired 6 employees recently and let X be the number of females in the group. Further suppose that $P(X = 1) = 0.1$, $P(X = 2) = 0.3$, $P(X = 3) = 0.5$, $P(X = 4) = 0.05$, $P(X = 5) = P(X = 6) = 0.025$
- The pmf of X is specified by $p(1) = 0.1$, $p(2) = 0.3$, $p(3) = 0.5$, $p(4) = 0.05$, $p(5) = p(6) = 0.025$, and $p(x) = 0$ for all other values of x

Random Variables

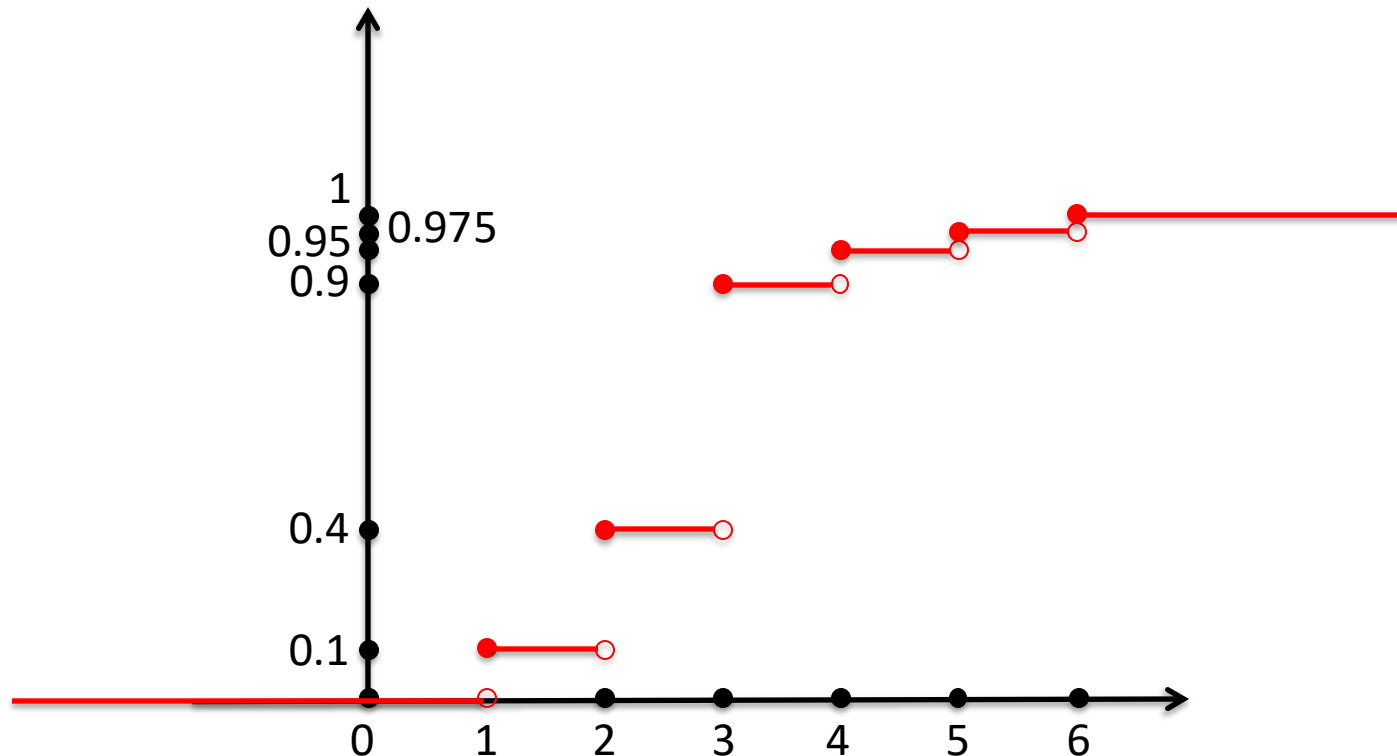
- Can also define the *cumulative distribution function* (cdf) of a random variable X as $F(x) = P(X \leq x) = \sum_{a \leq x} P(X = a) = \sum_{a \leq x} p(a)$



Random Variables

- Can also define the *cumulative distribution function* (cdf) of a random variable X as $F(x) = P(X \leq x) = \sum_{a \leq x} P(X = a) = \sum_{a \leq x} p(a)$
- The cdf of X in the previous example:

$F(x) = 0$ for $x < 1$, 0.1 for $1 \leq x < 2$, 0.4 for $2 \leq x < 3$, 0.9 for $3 \leq x < 4$, 0.95 for $4 \leq x < 5$, 0.975 for $5 \leq x < 6$, and 1 for $x \geq 6$



Random Variables

- The pmf/cdf of a random variable X specifies it completely
- Knowing pmf is equivalent to knowing cdf, which is equivalent to specifying X completely

Random Variables

- The *joint probability mass function* of two (discrete) random variables X and Y is $p(x,y) = P(X=x, Y=y)$
- Given the joint pmf of two (discrete) random variables X and Y is $p(x,y) = P(X=x, Y=y)$, the marginal pmfs are given by

$$p_X(x) = P(X=x) = \sum_y P(X=x, Y=y) = \sum_y p(x,y)$$

and

$$p_Y(y) = P(Y=y) = \sum_x P(X=x, Y=y) = \sum_x p(x,y)$$

Random Variables

- Definition: two discrete random variables X and Y are *independent* if for any x and y , $P(X=x, Y=y) = P(X=x)P(Y=y)$
- In other words, $p(x, y) = p_X(x)p_Y(y)$ for all x and y

Expectation of Random Variables

- Arguably the most important characteristic of a random variable is its *expected value/mean/expectation*
- Definition: the expected value of a discrete random variable X is defined by $E[X] = \sum_x xP(X = x) = \sum_x xp(x)$

Expectation of Random Variables

- **Example.** What is the expected value of the outcome of a fair die roll?

$$E[X] = 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + 3 \times \frac{1}{6} + \dots + 6 \times \frac{1}{6}$$

$$= 3.5$$

Expectation of Random Variables

- **Example.** What is the expected value of the outcome of a fair die roll?
- $E[X] = 1 \times 1/6 + 2 \times 1/6 + 3 \times 1/6 + 4 \times 1/6 + 5 \times 1/6 + 6 \times 1/6 = 7/2$

Expectation of Random Variables

- **Example.** What is the expected value of the outcome of a fair die roll?
- $E[X] = 1 \times 1/6 + 2 \times 1/6 + 3 \times 1/6 + 4 \times 1/6 + 5 \times 1/6 + 6 \times 1/6 = 7/2$
- **Example.** What is the expected value of the number of coin tosses until we see a Heads, for a biased coin of coming up Heads with probability 0.3?

$X = \# \text{ coin toss to see H}$

X takes values $1, 2, 3, \dots$

$$p(k) = (0.7)^{k-1} \cdot 0.3, \quad k = 1, 2, 3, \dots$$

$$E(X) = \sum_{k=1}^{\infty} k p(k)$$

Expectation of Random Variables

- **Example.** What is the expected value of the outcome of a fair die roll?
- $E[X] = 1 \times 1/6 + 2 \times 1/6 + 3 \times 1/6 + 4 \times 1/6 + 5 \times 1/6 + 6 \times 1/6 = 7/2$
- **Example.** What is the expected value of the number of coin tosses until we see a Heads, for a biased coin of coming up Heads with probability 0.3?
- Let X = the number of tosses until Heads. Then $P(X=1)=0.3$, $P(X=2)=0.7 \times 0.3$, ..., $P(X=k) = 0.7^{k-1} \times 0.3$, ...
- $E[X] = 1 \times P(X=1) + 2 \times P(X=2) + 3 \times P(X=3) + \dots = \sum_k k \times 0.7^{k-1} \times 0.3 = \dots = 1/0.3 = 10/3$

Expectation of Random Variables

- $E[X+Y] = E[X] + E[Y]$ is true in general
- Can generalize the argument to show that for discrete random variables $X_1, X_2, \dots, X_n$, $E[X_1+X_2+\dots+X_n] = E[X_1]+\dots+E[X_n]$
- **Example.** A construction firm has recently sent in bids for 3 jobs worth (in profits) 10, 20, and 40 thousand dollars respectively. If the probabilities of winning the jobs are respectively 0.2, 0.8 and 0.3, what is the firm's expected total profit?

X = total profit

$$E[X] = \sum_x x p(x)$$

$\nwarrow$ $\nearrow$ difficult

$0, 10, 20, 30, 40, 50, \dots$

X = total profit

$$= X_1 + X_2 + X_3$$

where X_i = profit from project i

$$E[X] = E[X_1] + E[X_2] + E[X_3] = 30$$

$$E[X_1] = 10 \times 0.2 + 0 \times 0.8 = 2$$

$$E[X_2] = 20 \times 0.8 = 16$$

$$E[X_3] = 40 \times 0.3 = 12$$

Expectation of Random Variables

- $E[X+Y] = E[X] + E[Y]$ is true in general
- Can generalize the argument to show that for discrete random variables $X_1, X_2, \dots, X_n$, $E[X_1+X_2+\dots+X_n] = E[X_1]+\dots+E[X_n]$
- **Example.** A construction firm has recently sent in bids for 3 jobs worth (in profits) 10, 20, and 40 thousand dollars respectively. If the probabilities of winning the jobs are respectively 0.2, 0.8 and 0.3, what is the firm's expected total profit?
- **Solution.** Let X_i , $i = 1, 2, 3$ denote the firm's profit from job i , then total profit = $X_1 + X_2 + X_3$. So $E[\text{total profit}] = E[X_1 + X_2 + X_3] = E[X_1] + E[X_2] + E[X_3]$.
- $E[X_1] = 10 \times 0.2 + 0 \times 0.8 = 2$; $E[X_2] = 16$ and $E[X_3] = 12$
- $E[X_1 + X_2 + X_3] = 2 + 16 + 12 = 30$

Expectation of Random Variables

- **Example.** What is the expected value of the number of times that in a thoroughly shuffled deck of 52 cards two adjacent cards are of the same rank (two aces, two kings, etc)?

X = # times that current card has same rank as previous card

$$= X_1 + \dots + X_{51}$$

$$X_i = \begin{cases} 1 & \text{if } i\text{-th card same rank as } (i+1)\text{-th card} \\ 0 & \text{otherwise} \end{cases}$$

↑
indicator variable

$$E[X] = E[X_1] + E[X_2] + \dots + E[X_n]$$

$$E[X_1] = 1 \times \frac{3}{51} + 0 \times \frac{48}{51} = \frac{3}{51}$$

$$= P(X_1 = 1) = P(\text{1st \& 2nd cards have same rank})$$

$$E[X_2] = \frac{3}{51} \qquad = \frac{3}{51} \times 51 = 3$$

$$\vdots$$

$$E[X_n] = \frac{3}{51}$$

Expectation of Random Variables

- **Example.** What is the expected value of the number of times that in a thoroughly shuffled deck of 52 cards two adjacent cards are of the same rank (two aces, two kings, etc)?
- **Solution.** Let X_i be 1 if the cards in position i and $i+1$ are of the same rank, and 0 otherwise. Then the quantity of interest is $E[X_1 + X_2 + \dots + X_{51}] = E[X_1] + \dots + E[X_{51}]$.
- $E[X_1] = 1 \times 3/51 + 0 \times 48/51 = 3/51$
- Similarly, $E[X_i] = 3/51$
- $E[X_1 + X_2 + \dots + X_{51}] = 3/51 \times 51 = 3$

Expectations of Functions of RV

- Often may be interested in expectation of a function of a random variable
- In general:

$$E[f(X)] = \sum_x f(x)P(X = x) = \sum_x f(x)p(x)$$

Expectations of Functions of RV

- Often may be interested in expectation of a function of a random variable
- In general:

$$E[f(X)] = \sum_x f(x)P(X = x) = \sum_x f(x)p(x)$$

- **Example.** R.v. X has the p.m.f $p(2) = 0.2$, $p(1) = 0.5$, $p(-1) = 0.3$. Compute $E[X^2]$.

$$E[X^2] = 2^2 \times 0.2 + 1^2 \times 0.5 + (-1)^2 \times 0.3$$

Expectations of Functions of RV

- Often may be interested in expectation of a function of a random variable

- In general:

$$E[f(X)] = \sum_x f(x)P(X = x) = \sum_x f(x)p(x)$$

- **Example.** R.v. X has the p.m.f $p(2) = 0.2$, $p(1) = 0.5$, $p(-1) = 0.3$. Compute $E[X^2]$.
- **Solution.** $E[X^2] = \sum_x x^2 P(X = x) = 2^2 P(X = 2) + 1^2 P(X = 1) + (-1)^2 P(X = -1) = 1.6$

Expectations of Functions of RV

- Also, for any constant a and b ,

$$E[aX + b] = aE[X] + b$$

- Question: is $E[f(X)] = f(E[X])$ in general? E.g., is $E[X^2] = (E[X])^2$?

No

No

Variance of Random Variables

- Write $\mu = E[X]$
- Definition: *Variance* of X , denoted by $\text{Var}(X)$, is defined to be
$$\text{Var}(X) = E[(X-\mu)^2]$$
- Equivalent formula for variance: $\text{Var}(X) = E[X^2] - (E[X])^2$
- Standard deviation = $\sqrt{\text{Var}(X)}$

Properties of Variance

- $\text{Var}(aX + b) = a^2\text{Var}(X)$
- Variance of a constant is 0
- Is it true that $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$?

generally false
but true when X & Y are
indep or uncorrelated

Covariance

- Definition: the *covariance* of two random variables X and Y is defined to be

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)],$$

where $\mu_X = E[X]$ and $\mu_Y = E[Y]$

- Covariance measures how two random variables tend to “move” together
- Useful properties:
 - $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$
 - $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
 - $\text{Cov}(X, X) = \text{Var}(X)$
 - $\text{Cov}(aX, Y) = a\text{Cov}(X, Y)$
 - $\text{Cov}(X + Z, Y) = \text{Cov}(X, Y) + \text{Cov}(Z, Y)$
 - $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$

Covariance

- Definition: *Correlation* between X and Y is

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

Corr is always between -1 and 1

1 → perfectly positively correlated

X & Y move together

-1 → .. negatively ..
.. .. away

Covariance

- If X and Y are independent, $\text{Cov}(X, Y) = 0$, and $E[XY] = E[X]E[Y]$
- If X and Y are independent, then $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$
- Question: X and Y are independent. What're the expectation and variance of $X - Y$?

$$E[X - Y] = E[X] - E[Y]$$

$$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y)$$

because $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(-Y)$

$$\begin{aligned} &= \text{Var}(X) + (-1)^2 \text{Var}(Y) \text{ by independence} \\ &= \text{Var}(X) + \text{Var}(Y) \end{aligned}$$

If X and Y are not necessarily
indep^t, then

$$E[X - Y] = E[X] - E[Y]$$

is still true

but $\text{var}(X - Y)$ may not be $\text{var}(X) + \text{var}(Y)$

Probability Density Function

- **Definition.** The random variable X is *continuous* if there exists a *nonnegative* function $f(x)$ (defined on the whole real line) so that $f(x)$ integrates to 1, i.e.,

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

and for every real number a ,

$$\mathbb{P}(X \leq a) = \int_{-\infty}^a f(x)dx$$

- In this case, $f(x)$ is called the *probability density function* (pdf) of the random variable X
- The function $F(a) = P(X \leq a)$ is called the *cumulative distribution function* (cdf) of X

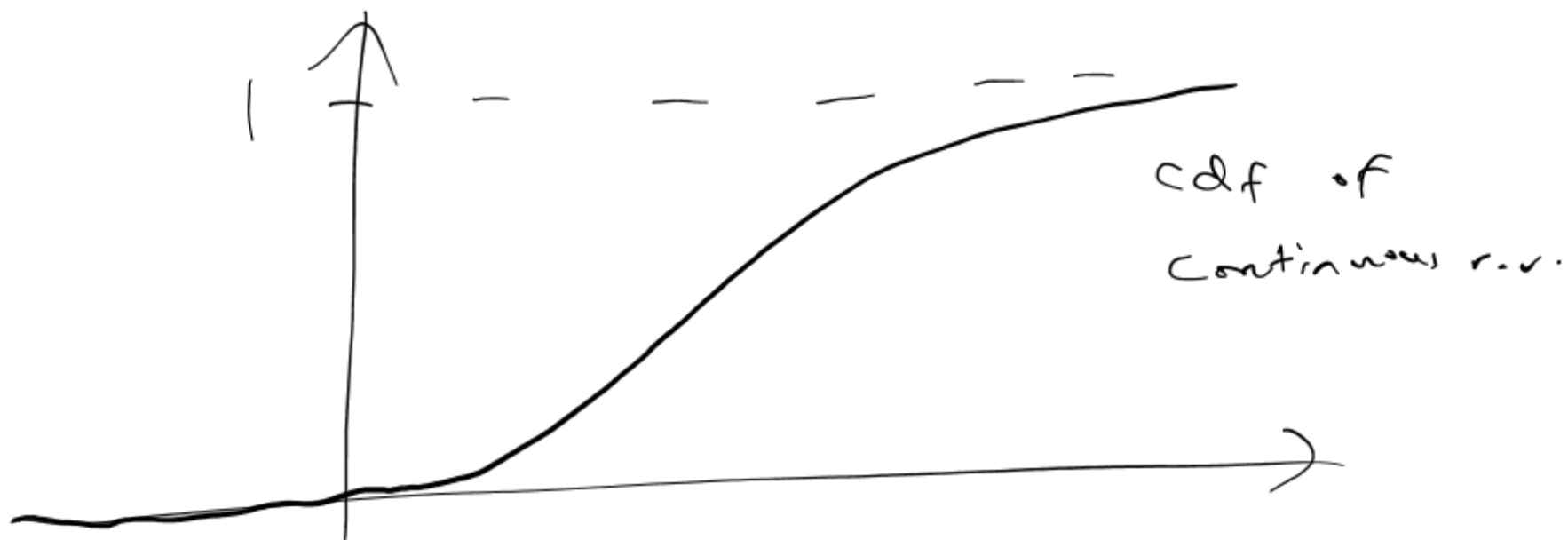
Some Discussion

- **Remark 1.** $f(x)$ is the derivative of $F(a) = P(X \leq a)$
- **Remark 2.** Since F is differentiable, it is a continuous function. In particular, it has no “jumps” like the cdf of discrete random variables
- **Remark 3.** $f(x)$ is NOT $P(X = x)$ ($= 0$)
- **Remark 4.** One way to make sense of $f(x)$ is as follows

$$\mathbb{P}(a < X \leq b) = \mathbb{P}(X \leq b) - \mathbb{P}(X \leq a) = \int_a^b f(x)dx$$

$$\mathbb{P}(x < X \leq x + \Delta x) = \int_x^{x+\Delta x} f(x)dx \approx f(x)\Delta x$$

for small Δx ; $f(x)$ itself is NOT a probability, but it does measure how likely it is to find a point near x



Joint Probability Density Function

- Similar to the discrete case, we can define joint distributions for continuous random variables
- **Definition.** Continuous random variables X and Y have a joint pdf $f(x, y)$ if $f(x, y) \geq 0$ for all x and y ,

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1,$$

and for all $-\infty < a, b < \infty$,

$$\mathbb{P}(X \leq a, Y \leq b) = \int_{-\infty}^b \int_{-\infty}^a f(x, y) dx dy$$

Independent Continuous RV

- **Definition (Marginal pdf).** Let X and Y be continuous random variables with joint pdf $f(x, y)$. The marginal pdf f_X of X is given by

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

- and the marginal pdf f_Y of Y is given by

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

- Can show that (similar for Y)

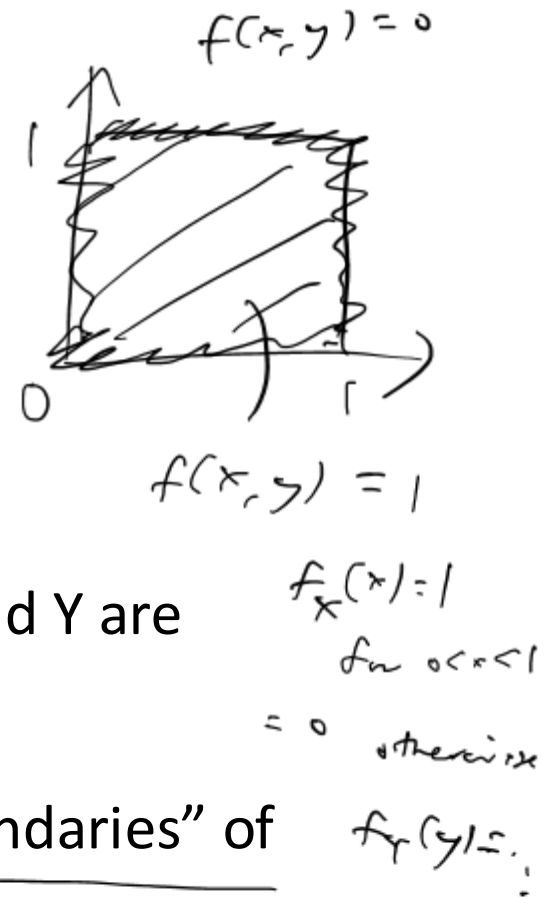
$$P(X \leq x) = \int_{-\infty}^x f_X(u) du$$

- **Definition (Independence).** Continuous rv X and Y are independent if for any x and y ,

$$P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$$

or equivalently, for any x and y not at the "boundaries" of discontinuity

$$f(x, y) = f_X(x)f_Y(y)$$



Expectation of Continuous RV

- Recall the definition of expectation for discrete rv
- Suppose X takes values $x_1, x_2, \dots$ and $P(X = x_i) = p_i$
- Then $E[X] = p_1x_1 + p_2x_2 + \dots$
- **Definition.** The *expectation* of a continuous random variable X with pdf $f(x)$ is defined by

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x) dx$$

- **Remark.** $f(x)dx$ serves as the continuous analog of p_i in the discrete case
- Variance is also defined the same as in the discrete case

Properties of Expectation and Variance

- Similarly, the covariance between two continuous r.v.'s X and Y is

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

- All properties of expectation, variance and covariance for discrete r.v.'s hold also for continuous r.v.'s. For example,
- $E[X+Y] = E[X] + E[Y]$
- $E[aX+b] = aE[X] + b$
- $\text{Var}(aX+b) = a^2\text{Var}(X)$
- $\text{Cov}(X, Y) = E[(X-\mu_X)(Y-\mu_Y)] = E[XY] - E[X]E[Y]$
- $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$
- If X and Y are independent, then $E[XY] = E[X]E[Y]$, $\text{Cov}(X, Y) = 0$, and $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$

Conditional Probabilities

- For a pair of joint discrete r.v.'s X and Y , with pmf $p(x, y)$, the conditional pmf of X given $Y=y$ is

$$p_{X|Y}(x|y) = P(X = x|Y = y) = \frac{p(x, y)}{p_Y(y)}$$

- The conditional cdf of X given $Y=y$ is

$$F_{X|Y}(x|y) = P(X \leq x|Y = y) = \sum_{a \leq x} p_{X|Y}(a|y) = \frac{\sum_{a \leq x} p(a, y)}{p_Y(y)}$$

Conditional Probabilities

- For a pair of joint continuous r.v.'s X and Y , with pdf $f(x, y)$, the conditional pdf of X given $Y=y$ is

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}$$

- The conditional cdf of X given $Y=y$ is

$$F_{X|Y}(x|y) = \int_{-\infty}^x f_{X|Y}(u|y) du = \frac{\int_{-\infty}^x f(u, y) du}{f_Y(y)}$$

• $P(Y=y) = 0$ for continuous r.v. Y .

• $f_Y(y)$ is a pdf, not probability

conditional pdf cannot be mathematically understood as conditional prob.

Conditional Probabilities

- For two r.v.'s X and Y , we have

$$E[Y] = E[E[Y|X]]$$

$E[Y|X]$ is
interpreted as
a r.v.

$$E[Y|X=\kappa]$$

This is the law of total probability / expectation or tower property

- Define $\text{Var}(Y|X) = E[(Y - E[Y|X])^2|X]$. We also have

$$\text{Var}(Y) = \text{Var}(E[Y|X]) + E[\text{Var}(Y|X)]$$

This is the variance decomposition or law of total variance

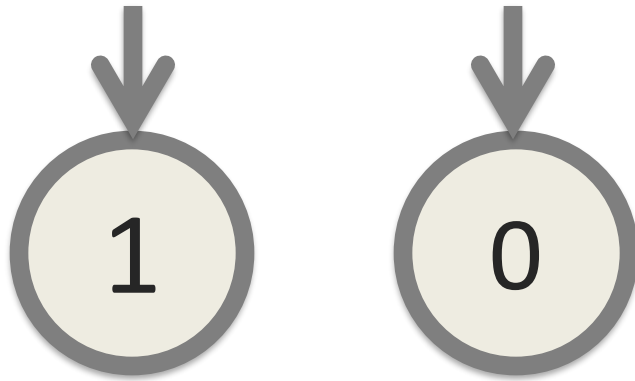
Discrete Uniform Random Variable

- A discrete uniform random variable X on $\{a, a+1, \dots, b\}$ is equally likely to be any x in $\{a, a+1, \dots, b\}$
- E.g., fair die roll; fair coin toss
- $P(X = x) = 1/(b-a+1)$, $x = a, a+1, \dots, b$

- $E[X]; \text{Var}(X)$
$$E[X] = \frac{1}{b-a+1} (a + (a+1) + \dots + b)$$
$$\text{Var}(X) = \frac{1}{b-a+1} \left((a - E[X])^2 + \dots + (b - E[X])^2 \right)$$

Bernoulli Random Variables

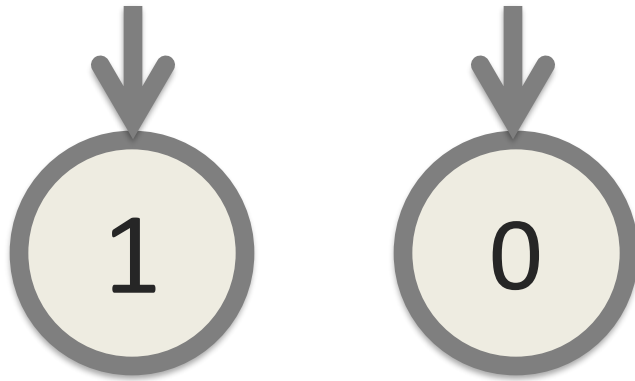
- A Bernoulli random variable is used to indicate the success/failure of an event, with a success probability p ; e.g., a coin toss
- Sometimes called Bernoulli trial
- $P(X = 1) = p$; $P(X = 0) = 1-p$



$$\begin{aligned} E[X] &= 1 \times P(X=1) + 0 \times P(X=0) = p \\ \text{var}(X) &= E[X^2] - (E[X])^2 = E[X] - (E[X])^2 = p - p^2 \\ &= p(1-p) \end{aligned}$$

Bernoulli Random Variables

- A Bernoulli random variable is used to indicate the success/failure of an event, with a success probability p ; e.g., a coin toss
- Sometimes called Bernoulli trial
- $P(X = 1) = p$; $P(X = 0) = 1-p$



- $E[X] = p$
- $\text{Var}(X) = p(1 - p)$
- Notation: $\text{Ber}(p)$

Binomial Distribution

- Consider n independent repetitions of a Bernoulli trial with success probability p
- X denotes the number of successes in n trials
- Pmf: $P(X = k) = p(k) = \binom{n}{k} p^k (1-p)^{n-k}$, $k=0, \dots, n$

$$X = X_1 + \dots + X_n$$

$$X_i \sim \text{Ber}(p) \quad \text{indep.}$$

$$E[X] = E[X_1] + \dots + E[X_n] = np$$

$$\text{Var}(X) = \text{Var}(X_1) + \dots + \text{Var}(X_n) = np(1-p)$$

Binomial Distribution

- Consider n independent repetitions of a Bernoulli trial with success probability p
- X denotes the number of successes in n trials
- Pmf: $P(X = k) = p(k) = \binom{n}{k} p^k (1-p)^{n-k}$, $k=0, \dots, n$
- $E[X] = np$
- $\text{Var}(X) = np(1-p)$
- Denoted by $\text{Bin}(n, p)$

Geometric Distribution

- $X = \#$ of Bernoulli trials with success probability p until we see a first success
- Pmf: $P(X = k) = (1-p)^{k-1}p$, $k = 1, 2, \dots$
- $E[X] = 1/p$
- $\text{Var}(X) = (1-p)/p^2$
- Denoted by $\text{Geo}(p)$

Geometric Distribution

- **Example.** Tom and his mom lost each other while wandering through a crowded amusement park. Their iphones are out of battery. Tom's mom, being a wise woman, foresaw this and made a prior agreement with Tom. Tom will wait at the entrance of one of the 15 attractions and his mom searches the attractions in a random order (the "wait-for-mommy" strategy). What are the expected value and standard deviation of the number of searches before Tom's mom finds him?

$X = \# \text{ searches before Tom's mom finds him}$

X is not Geo because it cannot be > 15 .

X is not bin because it cannot be 0.

X is uniform over $1, \dots, 15$



$$P(X=k) = \frac{1}{15} \quad \text{for } k = 1, \dots, 15$$

Geometric Distribution

- **Example.** Tom and his mom lost each other while wandering through a crowded amusement park. Their iphones are out of battery. Tom's mom, being a wise woman, foresaw this and made a prior agreement with Tom. Tom will wait at the entrance of one of the 15 attractions and his mom searches the attractions in a random order (the “wait-for-mommy” strategy). What are the expected value and standard deviation of the number of searches before Tom's mom finds him?
- Suppose now that Tom and his mom agrees that, in intervals of 10 minutes, they will each pick a random entrance and go to their chosen entrances. What are the expected value and standard deviation of the number of searches in this case?

$$X \sim \text{Geo}\left(\frac{1}{15}\right)$$

Poisson Distribution

- Poisson r.v. counts the number of occurrences of interest in a segment of time or space; possible values are 0, 1, 2, ...
- Poisson distribution with parameter $\lambda = np$ approximates $\text{Bin}(n, p)$ when n is large and p is small; e.g., number of calls received by an emergency center
- X has a Poisson distribution with parameter $\lambda > 0$ if its pmf satisfies

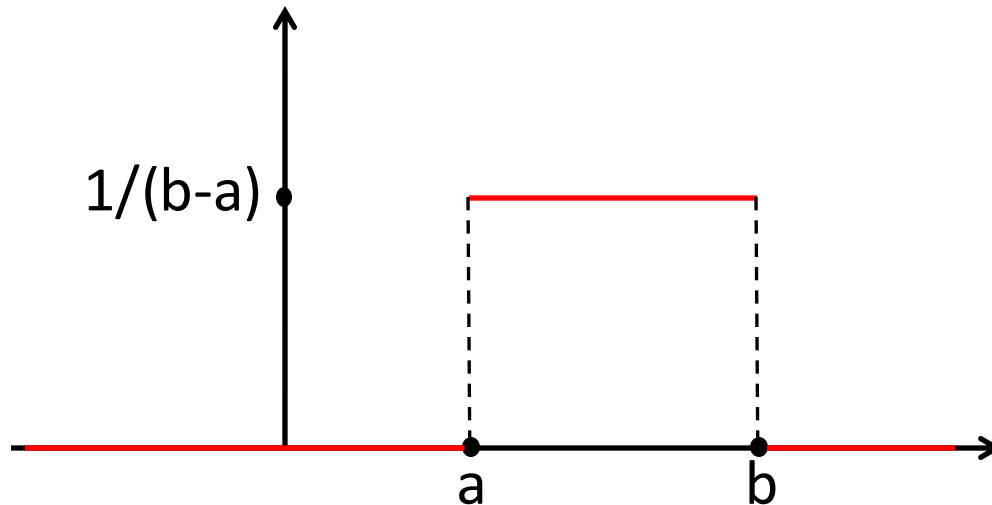
$$P(X = k) = e^{-\lambda} \lambda^k / k!, \quad k = 0, 1, 2, \dots$$

- $E[X] = \text{Var}(X) = \lambda$
- Denoted by $\text{Poi}(\lambda)$

Uniform Random Variables

- A random variable X is uniformly distributed over $[a, b]$ if its probability density function (pdf) $f(x)$ is

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$



- Denote it by $\text{Unif}(a, b)$
- Used to model a “completely random” selection over the interval

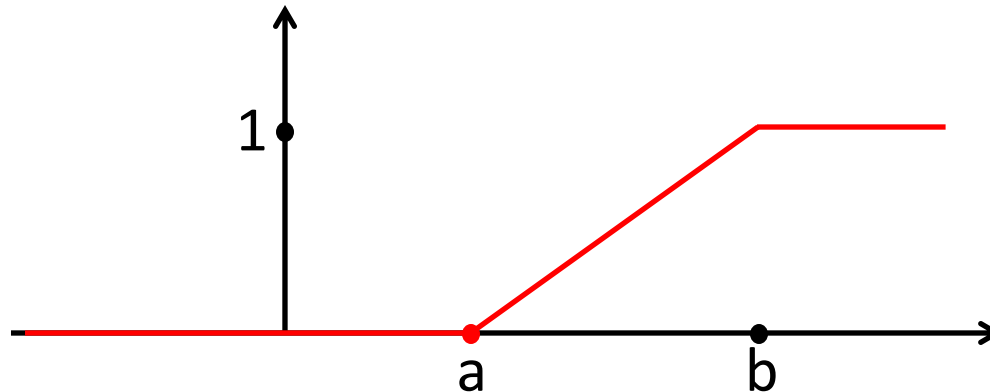
Uniform Random Variables

- Note that this is a valid pdf because

$$\int_a^b \frac{1}{b-a} dx = \frac{1}{b-a} \int_a^b dx = \frac{1}{b-a} \cdot (b-a) = 1$$

- The (cumulative) distribution function of $\text{Unif}(a, b)$ is

$$F(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } a \leq x \leq b \\ 1 & \text{if } x > b. \end{cases}$$



Uniform Random Variables

- Expectation of $X \sim \text{Unif}(a, b)$ is

$$\mathbb{E}[X] = \int_a^b x \cdot \frac{1}{b-a} dx = \left. \frac{x^2}{2(b-a)} \right|_a^b = \frac{a+b}{2}$$

- Variance of X can be computed as follows

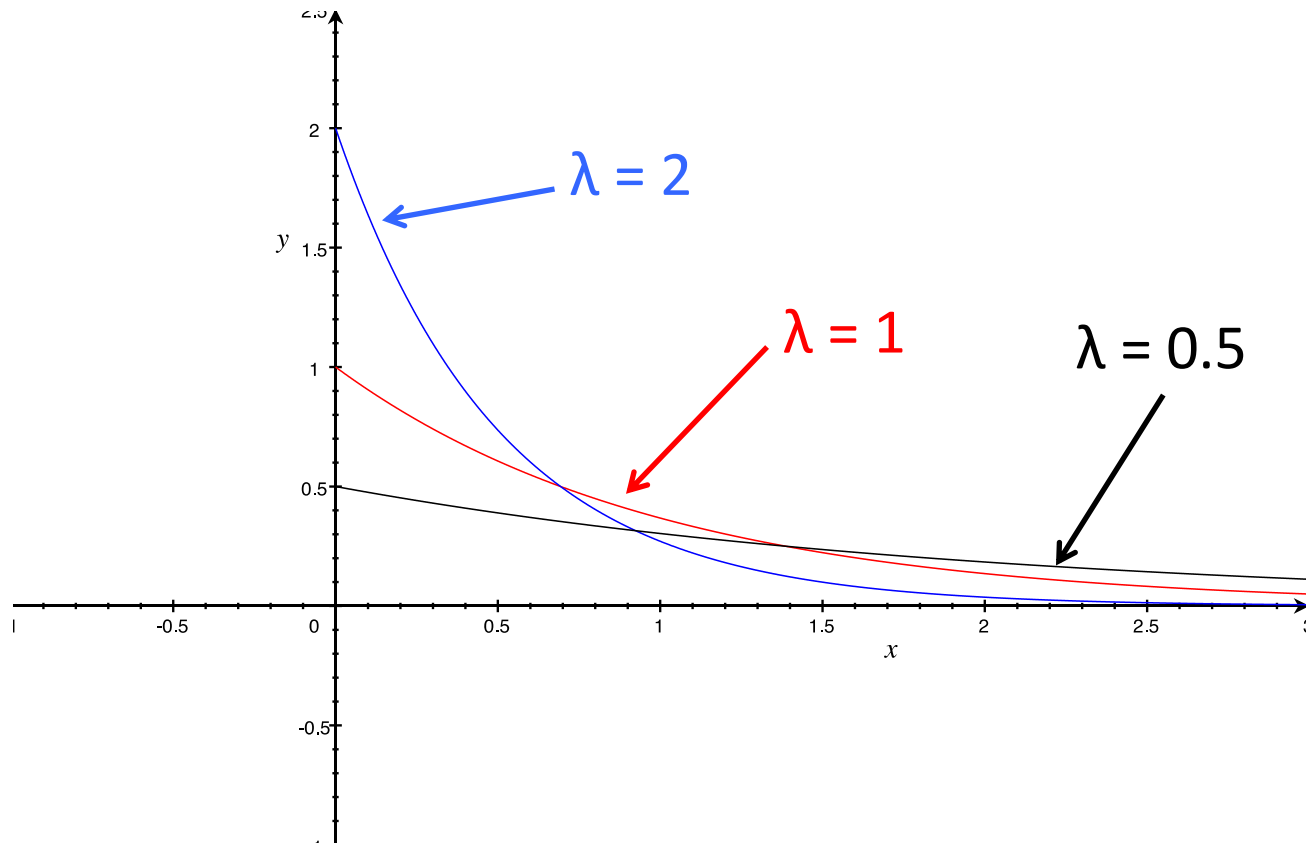
$$\mathbb{E}[X^2] = \int_a^b x^2 \cdot \frac{1}{b-a} dx = \left. \frac{x^3}{3(b-a)} \right|_a^b = \frac{b^3 - a^3}{3(b-a)}$$

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{b^3 - a^3}{3(b-a)} - \left(\frac{a+b}{2} \right)^2 \\ &= \frac{b^2 + ab + a^2}{3} - \frac{a^2 + 2ab + b^2}{4} = \frac{(b-a)^2}{12} \end{aligned}$$

Exponential Random Variables

- A random variable X has an *exponential* pdf with parameter $\lambda > 0$ if the pdf $f(x)$ is

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

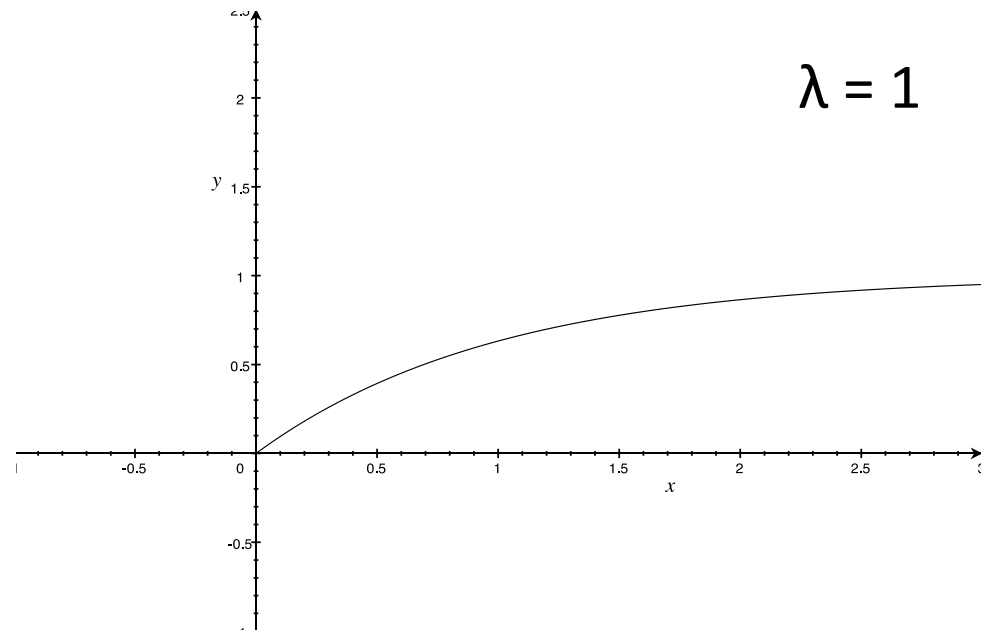


Exponential Random Variables

- Denoted by $\text{Exp}(\lambda)$
- Often used to model the time till some event (e.g., earthquake, phone call, radioactive decay, etc) occurs
- cdf $F(x)$ is given by

$$F(x) = \int_0^x \lambda e^{-\lambda y} dy = -e^{-\lambda y} \Big|_0^x = 1 - e^{-\lambda x}, x \geq 0$$

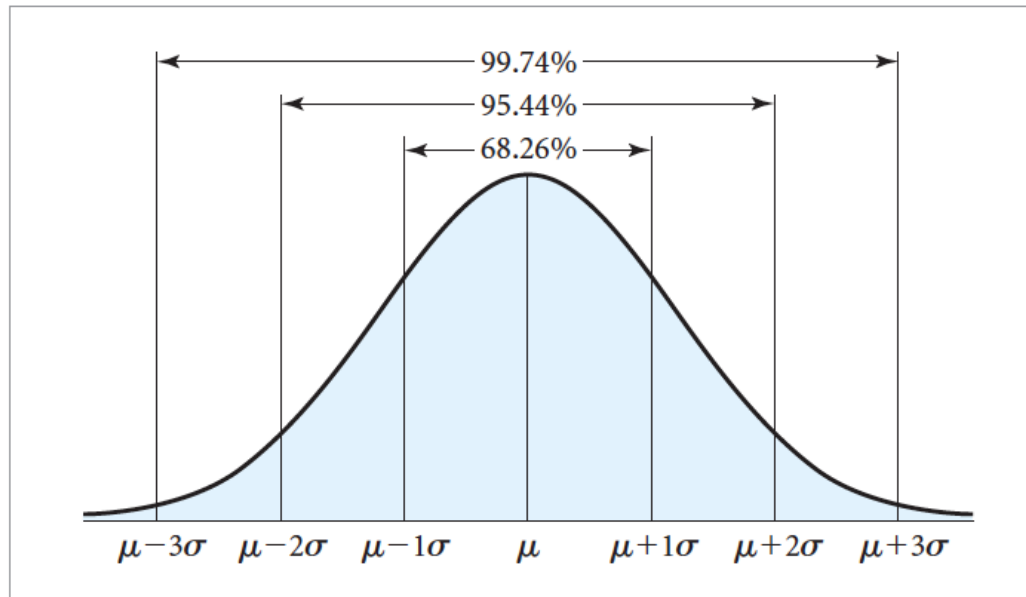
- $E[X] = 1/\lambda$
- $\text{Var}(X) = 1/\lambda^2$



Normal Random Variables

- A bell-shaped pdf curve
- Random variable X is normally distributed with mean μ and variance σ^2 if it has pdf

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, x \in \mathbb{R}$$



Normal Random Variables

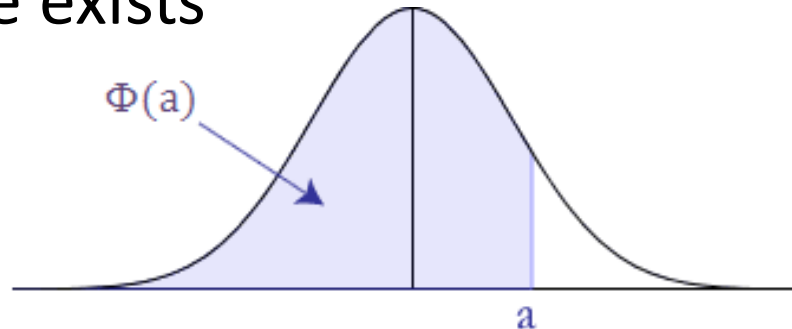
- Normal distribution and r.v. show up in many natural phenomena and are important in statistics (later)
- $E[X] = \mu$, $\text{Var}(X) = \sigma^2$
- Denote it by $N(\mu, \sigma^2)$
- $N(0, 1)$ is called the standard normal distribution

Normal Random Variables

- **Property 1:** If $X \sim N(\mu, \sigma^2)$, then $aX+b \sim N(a\mu+b, a^2\sigma^2)$
- **Property 1':** If $X \sim N(\mu, \sigma^2)$, then $Z = (X-\mu)/\sigma \sim N(0, 1)$
- Moral: all normal rv can be “reduced” to the standard normal
- Often use $\Phi(x)$ to denote the cdf of standard normal

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy$$

- No closed-form formula for $\Phi(x)$ but the standard normal table exists



use property 1
with
 $a = \frac{1}{\sigma}, b = -\frac{\mu}{\sigma}$

Normal Random Variables

- **Property 2:** if $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ are independent normal rv, then $X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$